

Visualization and Reporting

```
In [1]: import pandas as pd
df = pd.read_csv('ClusteredData.csv')
```

Purchase categories

```
In [2]: df_categories = df.copy()
```

```
In [3]: # Splitting delimiter is ;
df_categories['Purchase_Categories'] =
df_categories['Purchase_Categories'].str.split(';')
```

```
In [4]: # Splitting gave us a list of separated categories
df_categories['Purchase_Categories'].head()
```

Out[4]:

	Purchase_Categories
0	[Beauty and Personal Care]
1	[Beauty and Personal Care, Clothing and Fashio...
2	[Groceries and Gourmet Food]
3	[Groceries and Gourmet Food, Beauty and Person...
4	[Groceries and Gourmet Food, Beauty and Person...

dtype: object

```
In [5]: # Splitting row wise for each category in the list
df_categories = df_categories.explode('Purchase_Categories')
```

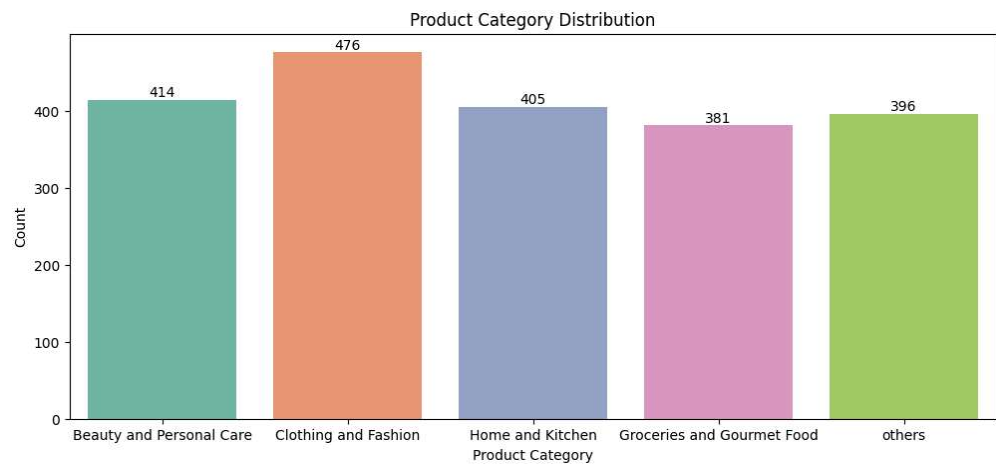
```
In [6]: # Clothing and Fashion is the Most Popular Product Category
# SnapDeal can do inventory planning prioritizing this category
df_categories['Purchase_Categories'].value_counts(normalize=True).round(4)*
100
```

Out[6]:

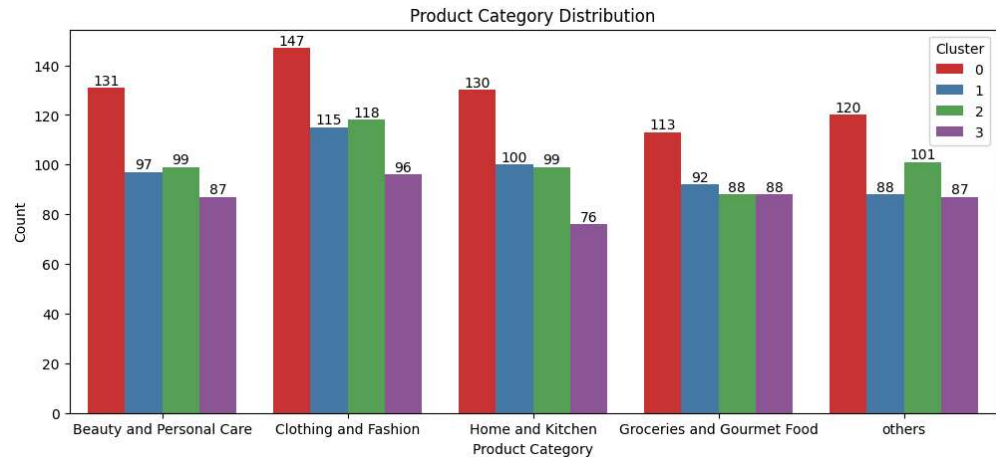
	proportion
Purchase_Categories	
Clothing and Fashion	22.97
Beauty and Personal Care	19.98
Home and Kitchen	19.55
others	19.11
Groceries and Gourmet Food	18.39

dtype: float64

```
In [8]: import matplotlib.pyplot as plt
import seaborn as sns
fig = plt.figure(figsize=(12,5))
ax = sns.countplot(data=df_categories,x='Purchase_Categories',
hue='Purchase_Categories', palette='Set2')
for p in ax.containers:
    ax.bar_label(p)
plt.xlabel('Product Category')
plt.ylabel('Count')
plt.title('Product Category Distribution')
plt.show()
```



```
In [27]: fig = plt.figure(figsize=(12,5))
ax = sns.countplot(data=df_categories,x='Purchase_Categories',
hue='Cluster', palette='Set1')
for p in ax.containers:
    ax.bar_label(p)
plt.xlabel('Product Category')
plt.ylabel('Count')
plt.title('Product Category Distribution')
plt.show()
```



Browsing frequency distribution

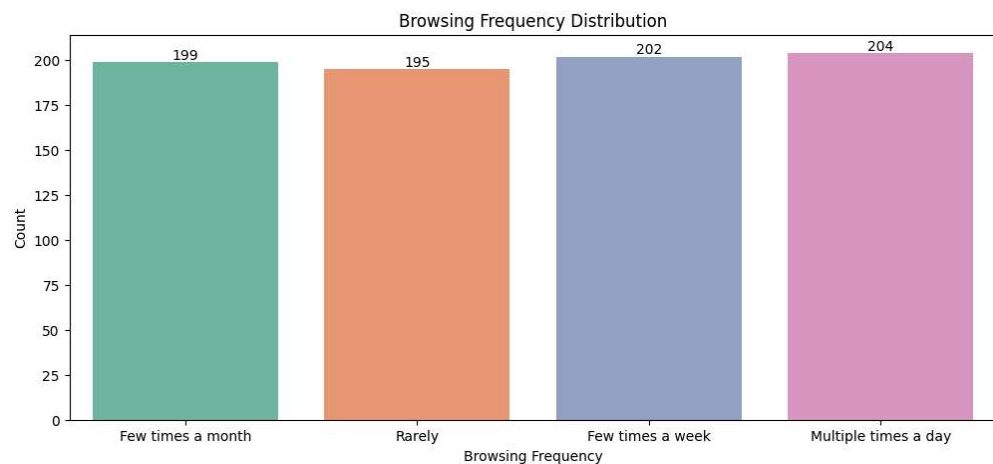
```
In [13]: df['Browsing_Frequency'].value_counts(normalize=True).round(4)*100
```

Out[13]:

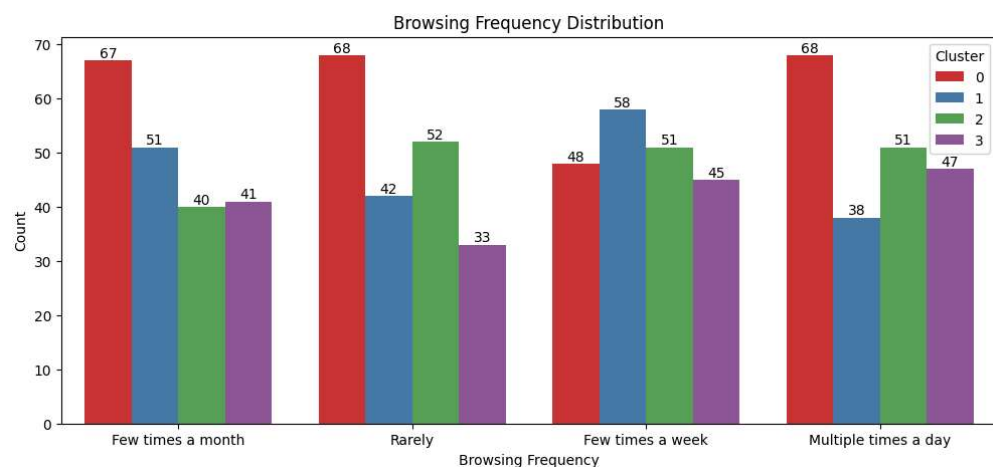
	proportion
Browsing_Frequency	
Multiple times a day	25.50
Few times a week	25.25
Few times a month	24.88
Rarely	24.38

dtype: float64

```
In [16]: # More than 50% of Customers are browsing everyday or few times a week
fig = plt.figure(figsize=(12,5))
ax = sns.countplot(data=df,x='Browsing_Frequency',
hue='Browsing_Frequency', palette='Set2')
for p in ax.containers:
    ax.bar_label(p)
plt.xlabel('Browsing Frequency')
plt.ylabel('Count')
plt.title('Browsing Frequency Distribution')
plt.show()
```



```
In [24]: # Cluster 0 customers are dominantly browsing multiple times a day than
others
fig = plt.figure(figsize=(12,5))
ax = sns.countplot(data=df,x='Browsing_Frequency', hue='Cluster',
palette='Set1')
for p in ax.containers:
    ax.bar_label(p)
plt.xlabel('Browsing Frequency')
plt.ylabel('Count')
plt.title('Browsing Frequency Distribution')
plt.show()
```



Satisfaction levels

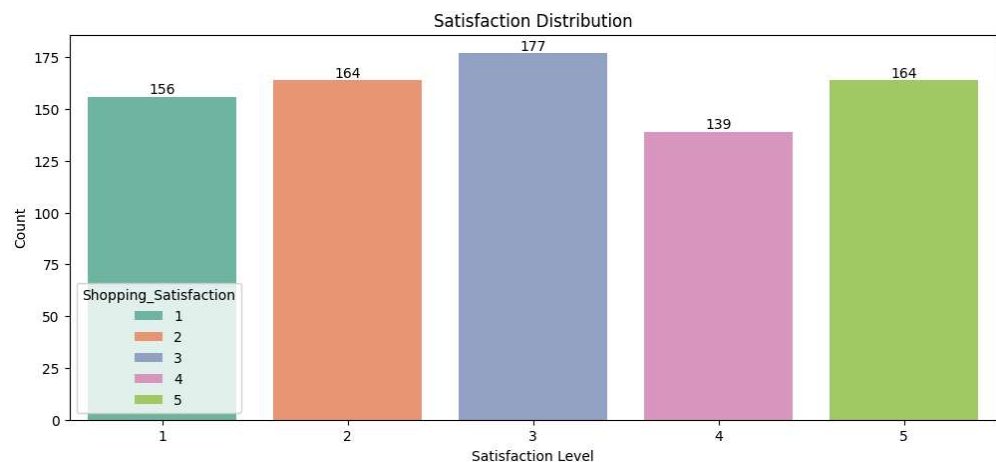
```
In [19]: # Most customers lie at satisfaction level 3 out of 5
df['Shopping_Satisfaction'].value_counts(normalize=True).round(4)*100
```

Out[19]:

	proportion
Shopping_Satisfaction	
3	22.12
2	20.50
5	20.50
1	19.50
4	17.37

dtype: float64

```
In [20]: fig = plt.figure(figsize=(12,5))
ax = sns.countplot(data=df,x='Shopping_Satisfaction',
hue='Shopping_Satisfaction', palette='Set2')
for p in ax.containers:
    ax.bar_label(p)
plt.xlabel('Satisfaction Level')
plt.ylabel('Count')
plt.title('Satisfaction Distribution')
plt.show()
```



Correlation between recommendation helpfulness and satisfaction

```
In [31]: rec = {'Yes': 3, 'No': 1, 'Sometimes': 2}
df['Recommendation_Helpfulness'] =
df['Recommendation_Helpfulness'].map(rec)
```

```
In [33]: # Almost negligible, slight positive Correlation
c = df[['Recommendation_Helpfulness', 'Shopping_Satisfaction']].corr()
c
```

Out[33]:

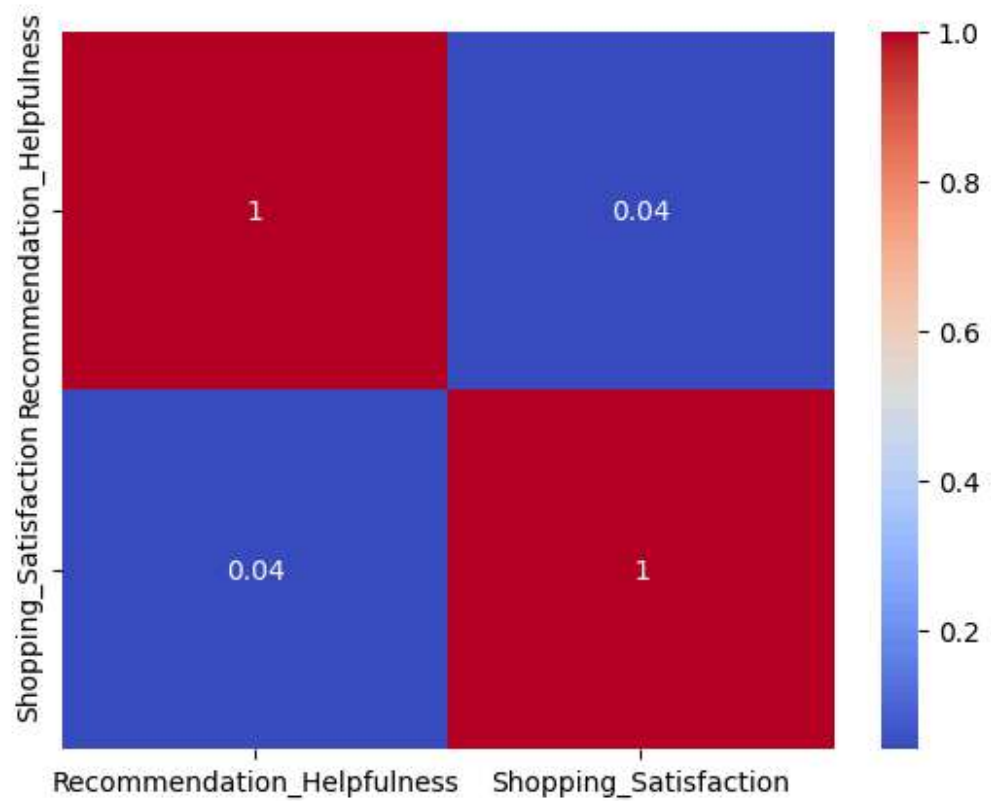
	Recommendation_Helpfulness	Shopping_Satis
Recommendation_Helpfulness	1.000000	0.040151
Shopping_Satisfaction	0.040151	1.000000

```
In [39]: # Most customers fall in the category of NO helpfulness of the
recommendations
# And their avg Satisfaction is also Low
# More Personalized and quality recommendation is required.
df.groupby('Recommendation_Helpfulness').agg({'Shopping_Satisfaction':
['mean', 'count']})
```

Out[39]:

	Shopping_Satisfaction	
	mean	count
Recommendation_Helpfulness		
1	2.888136	295
2	3.077869	244
3	3.019157	261

```
In [40]: sns.heatmap(c, annot=True, cmap='coolwarm');
```



```
In [ ]:
```